ANATOMY OF AI AGENTS 2026

www.arianprabowo.com · 10 Stages (1. Silicon → 10. Multi-Agent) × 4 Organisms (LLM · RAG · Multimodal · Embodied AI)

What actually runs under an AI agent in 2026? Trace left to right:
1. Silicon → 2. Python → 3. Data → 4. Pre-Training → 5. SFT →
6. Post-Training → 7. Test-Time → 8. Context → 9. Harness → 10. Multi-Agent.

① COLUMNS (1 → 10): Physical Silicon to Multi-Agent Swarms
② ROWS (4 ORGANISMS): LLM Spine + RAG, Multimodal & Embodied AI
③ INSIDE EVERY CELL: Stack + Red ⚠ ANTI-PATTERNS Box at Bottom

*Synthesized from Haggai Roitman's 'The Hitchhiker's Guide to Agentic AI: From Foundations to Systems' (arXiv:2606.24937, v1.4, June 2026) + 2026 systems tooling. Companion to AI Skill Tree 2026 & AI Ladder Evolution by Arian Prabowo · Updated: 2026-11-08 01:38:44 AEDT (UTC+11:00)

ORGANISMS × STAGES 1. Silicon → 10. Swarms LLM · RAG · Multimodal · Embodied (+ ⚠ Traps)	**1. Silicon** Physical Compute Silicon, GPU Clusters, HBM3e Memory & Interconnect Fabrics GPUs/TPUs/NPUs, HBM3e vs. SRAM hierarchy, CUDA/Triton kernels, and NVLink/InfiniBand fabrics. • Core Unit: FLOPs, HBM Bytes & CUDA Warps • Guide: Ch. 1.6, 2.3, 11.10–14, 30.5 (GPUs & Kernels)	**2. Python** Python Runtimes, Package Managers, Virtual Environments & Tensor Compilers Package managers (Homebrew, uv, conda, pip, venv), Python 3.12+, PyTorch, JAX, vLLM, and sandboxes. • Core Unit: Wheels, Lockfiles, Tensors & Paged KV • Guide: Ch. 1.24–25, 11.1–9, 21.5, 22.1–9 (Env & Runtime)
CORE SPINE **LLM** Autoregressive Text, Code & Reasoning Language Models Token-by-Token Language, Symbolic Math, Code Gen & Autonomous Agent Loops • 1–3: GPU / PyTorch / BPE • 4–7: MoE / SFT / RLVR / CoT • 8–10: MCP / Harness^2 / A2A	NVIDIA H100 / H200 / Blackwell B200 HBM3e vs. On-Chip SRAM Hierarchy FlashAttention-3 IO-Aware SRAM Tiling Google TPU v5p / v6e Trillium / v7 CUDA · OpenAI Triton · CUTLASS FP8 (E4M3/E5M2) & Microscaling FP4 NVLink 5 (1.8 TB/s) & NVSwitch Model FLOPs Utilization (MFU > 45%) Prefill (Compute) vs. Decode (BW) Streaming Multiprocessors (132 SMs) 400G/800G RoCEv2 & InfiniBand NDR Confidential Compute GPU TEEs ⚠ ANTI-PATTERNS & FAILURE MODES × HBM Memory Bandwidth Wall (Decode) × CUDA Out-of-Memory (OOM) Crashes × Silent Tensor Core SDC Bit-Flips × Straggler GPU Halting 1,000 Nodes	Homebrew (brew) · uv · conda · pip · venv Python 3.12+ · PyTorch 2.x · JAX · MLX vLLM · SGLang · llama.cpp · Docker · E2B torch.compile (TorchInductor) & XLA PagedAttention KV & Continuous Batch pyproject.toml & uv.lock Reproducibility E2B / Firecracker MicroVM Sandboxes OS Sandboxing (Bubblewrap / Seatbelt) TTFT / TPOT SLAs & Asyncio Event Loop manylinux Wheels & CUDA ABI Tags Speculative Decoding Draft-Verify Outlines / XGrammar FSM Masking ⚠ ANTI-PATTERNS & FAILURE MODES × Python Dependency & CUDA ABI Hell × Unsandboxed Host RCE & rm -rf / × KV-Cache Fragmentation & Preemption × pip install Breaking flash-attn ABI
ORGANISM · KNOWLEDGE **RAG** Hybrid Vector, Lexical & Graph Retrieval-Augmented Generation External Grounding, Dense + BM25 Search, GraphRAG & Citation Verification • 1–3: SIMD / FAISS / Docling • 4–7: BGE-M3 / Rerank / HyDE • 8–10: Qdrant / Agentic RAG	NVIDIA cuVS / RAFT GPU Vector Search AVX-512 SIMD Distance Kernels NVMe Gen5 DirectStorage DiskANN I/O GPU-Accelerated CAGRA Graph Index Int8 / Binary Scalar & PQ Hardware High-IOPS NVMe SSD Page Cache RDMA Zero-Copy Index Sharding Sub-10ms p99 Vector Query SLA DRAM vs. SSD Tiered Memory Pool L2 / Cosine / Inner-Product SIMD GPUDirect Storage (GDS) DMA HNSW Graph Traversal Cache Misses ⚠ ANTI-PATTERNS & FAILURE MODES × Random-Read SSD IOPS Thrashing × HNSW Graph Exceeding Host DRAM × PCIe Host-to-GPU Index Bottleneck × NUMA Cross-Socket Latency Spikes	FAISS · HNSWlib · LanceDB · PyArrow psycopg3 (pgvector) · qdrant-client Apache Arrow Zero-Copy Memory Tables rank-bm25 · tantivy · Pyserini Lexical Parquet Columnar Storage & DuckDB Async gRPC / HTTP/2 Connection Pools ONNX Runtime / TensorRT Embed Serving Memory-Mapped (mmap) Vector Segments Connection Pool & Batch Upsert SLAs pyarrow.Table Zero-Copy Slicing SQLAlchemy / AsyncPG Vector Types Protobuf / FlatBuffers Serialization ⚠ ANTI-PATTERNS & FAILURE MODES × GIL Lockup During CPU Token/Chunking × Pickle Deserialization RCE in Indexes × Unbounded Connection Pool Exhaustion × Pandas Copy-on-Write RAM Explosion
ORGANISM · SENSORY **Multimodal** Vision, Audio, Speech, Video & Native Omni-Modal Models ViT Patch Perception, Native Speech Codecs, Spatio-Temporal Video & Cross-Modal Omni • 1–3: NVDEC / FFmpeg / Pairs • 4–7: SigLIP / LLaVA / Visual CoT • 8–10: Token Merge / LiveKit	Hardware NVDEC / NVENC Video Codecs ViT Patch Tensor-Core SRAM Tiling Dedicated Audio DSP & Neural Engines Apple Silicon M4 Unified Memory JPEG/HEVC Hardware Decode Pipelines 2D/3D Conv & DiT Attention Kernels GPUDirect RDMA Camera/Video Ingest Sub-200ms Voice-to-Voice Hardware SLA 4K Frame Buffer Bandwidth Budgeting NVMM Zero-Copy Frame Buffers BFloat16 ViT Patch Projection Hardware Acoustic Echo Cancellation ⚠ ANTI-PATTERNS & FAILURE MODES × CPU-Bound Video Decode Starving GPU × 4K Uncompressed Frame PCIe Saturation × Audio Codec Sample-Rate Clock Drift × Thermal Throttling on Continuous Video	torchvision · torchaudio · OpenCV FFmpeg · PyAV · Pillow-SIMD · Decord WebRTC · LiveKit · whisper.cpp · Kokoro NVIDIA DALI GPU Data Loading Pipeline librosa · soundfile · Soxr Resampling vLLM Multimodal & SGLang Vision Engine WebSockets / RTP Real-Time Audio Streams VAD (Silero) Turn-Detection Runtime Frame-Sync Jitter Buffer Management BGR-to-RGB Tensor Channel Ordering Opus / PCM16 Streaming Audio Frames EXIF Orientation Tag Normalization ⚠ ANTI-PATTERNS & FAILURE MODES × OpenCV BGR vs. RGB Silent Channel Bug × FFmpeg Zombie Subprocess Memory Leaks × Audio Buffer Underrun Pops & Glitches × Pillow Decompression Bomb DoS
ORGANISM · ACTION **Embodied AI** GUI Computer Use, Physics World Models & VLA Robotics Browser/OS GUI Control, 3D Physics Simulators, World Models & VLA Robot Policies • 1–3: Edge NPU / Isaac / Teleop • 4–7: VLA π0 / ACT / Sim2Real • 8–10: 3DGS / OSWorld / Fleets	NVIDIA Jetson Thor / Orin Edge SoCs GPU PhysX / Warp Parallel Sim Kernels 1 kHz Real-Time Deterministic Bus LiDAR / Depth / IMU Sensor ISP Silicon On-Device INT4/FP8 NPU Inference EtherCAT / CAN-FD Actuator Bus Hardware Watchdog & E-Stop Relays 10,000x Parallel Sim Env Throughput Sub-5ms Sensor-to-Actuator Loop Latency PREEMPT_RT Real-Time Linux Kernel MIPI CSI-2 Multi-Camera Deserializers Battery Watt-Hour Compute Envelopes ⚠ ANTI-PATTERNS & FAILURE MODES × Edge Thermal Throttling Mid-Trajectory × GC / OS Jitter Breaking 1 kHz Control × USB/CSI Camera Frame Drop Desync × Sim-to-Real Contact Physics Mismatch	ROS 2 (rclpy) · Gymnasium · MuJoCo NVIDIA Isaac Lab · ManiSkill · Genesis Playwright · CDP · PyAutoGUI · LeRobot ZeroMQ / CycloneDDS Pub-Sub Middleware Headless Chromium / Xvfb Virtual Displays Open3D · Trimesh · PyBullet Kinematics Dockerized Browser & OS Sandboxes Vectorized AsyncGym Env Stepping Deterministic Seed & Replay Recording URDF / MJCF Robot Description XML Accessibility Tree (AXTree) Snapshots 6-DoF End-Effector Quaternion Math ⚠ ANTI-PATTERNS & FAILURE MODES × ROS 2 DDS Discovery Multicast Storms × Zombie Headless Browser RAM Leaks × Euler Angle Gimbal Lock in Control × Non-Deterministic Physics Step Drift

ORGANISMS × STAGES	**3. Data** Web & Code Corpora, Subword Tokenization & Curation Pipelines Pre-training/post-training corpora, BPE/Tiktoken tokenizers, deduplication, filtering, and synthetic data. • Core Unit: Corpora, Subword Tokens & Shards • Guide: Ch. 1.2, 3.1, 3.3–3.5, 4.1, 30.2 (Data & Tokenizers)	**4. Pre-Training** Foundation Model Architectures, Sparse MoE Routing & Self-Supervised Pre-Training Transformer/MoE/hybrid architectures, attention (GQA, MLA), RoPE, next-token pre-training, and scaling laws. • Core Unit: Transformer Blocks, MoE & Base Weights • Guide: Ch. 1.1, 2.2, 3.1, 11.1–9, 30.1 (Architecture & Pre-Training)
LLM	FineWeb-Edu · Dolma · The Stack v2 Byte-Level BPE · Tiktoken · SentencePiece MinHash LSH Dedup & Quality Classifiers HF Datasets · Datatrove · NeMo Curator Execution-Verified Code & Math Traces Benchmark Decontamination (13-Gram) PII, Secret & License Scrubbing Vocabulary Size (128k–256k) & Fertility Domain Curriculum & Annealing Mixes cl100k / o200k Pre-Tokenization Regex Canary String Injection & Filtering Vendi Score & N-Gram Diversity ⚠ ANTI-PATTERNS & FAILURE MODES × Garbage-In Synthetic Model Collapse × Test-Set Contamination Mirage × Tokenizer Glitch Tokens & Fertility Bloat × PII & API Keys Baked into Weights	Transformer · Sparse MoE · GQA / MLA Megatron-LM · DeepSpeed · TorchTitan Chinchilla & Inference-Aware Scaling Laws Llama 4 · Qwen 3 · DeepSeek-V3 Base Hybrid Linear Attention (Mamba/DeltaNet) Pre-Norm RMSNorm, SwiGLU & RoPE/YaRN 3D Parallelism (DP/FSDP + TP + PP + EP) Loss-Spike Recovery & Z-Loss Stability Base Perplexity (Bits-per-Byte) & NIAH Aux-Loss-Free MoE Load Balancing muP Maximal Update Parametrization Async Distributed Checkpoint Shards ⚠ ANTI-PATTERNS & FAILURE MODES × Catastrophic Loss Spikes & NaN Divergence × MoE Routing Collapse (Expert Starvation) × Pipeline Bubble & All-to-All Bottlenecks × RoPE Extrapolation Collapse Past Train Len
RAG	Docling · Marker · Unstructured PDF ETL Semantic, Contextual & AST Code Chunking Metadata Schema, ACLs & Temporal Tags Anthropic Contextual Chunk Headers Table-to-Markdown & Layout Extraction Hard-Negative Triplet Mining (Q, d+, d-) Incremental CDC Document Sync Near-Duplicate Chunk Deduplication Golden QA Evaluation Set Curation 256-512 Token Overlap Sliding Windows Tree-sitter AST Function Boundary Split SHA-256 Content-Addressable Chunk IDs ⚠ ANTI-PATTERNS & FAILURE MODES × Mid-Sentence / Mid-Table Naive Chunking × Multi-Column PDF Reading-Order Scramble × Stale Deleted Docs Remaining in Index × Missing Document-Level RBAC Metadata	Bi-Encoder Contrastive InfoNCE Training BGE-M3 · NV-Embed-v2 · Cohere Embed v4 ColBERTv2 Late-Interaction MaxSim Matryoshka Representation Learning (MRL) SPLADE Learned Sparse Lexical Expansion In-Batch & Cross-GPU Negative Scaling Bidirectional Attention Encoder Backbones Multilingual Cross-Lingual Alignment MTEB Retrieval & NDCG@10 Benchmarks Temperature-Scaled Cosine Similarity Dimension Truncation (1024 -> 256d) Asymmetric Query/Passage Prefixes ⚠ ANTI-PATTERNS & FAILURE MODES × Embedding Space Anisotropy Collapse × False-Negative Collisions in Large Batches × Out-of-Domain Lexical Acronym Blindness × Fixed 512-Token Encoder Context Truncation
Multimodal	DataComp · LAION · OBELICS Interleaved OCR Document, Chart & UI Grounding Sets Whisper ASR & Diarized Speech Corpora CLIPScore & Aesthetic Quality Filtering Synthetic VLM Recaptioning (ShareGPT4V) Scene-Cut Video Clip Segmentation Discrete Audio Codec Tokens (EnCodec/Mimi) Face Blurring & C2PA Provenance Scrub Bounding-Box & Set-of-Mark Annotations Dynamic Aspect-Ratio AnyRes Tiling Forced Phoneme-Timestamp Alignment NSFW & Watermark Classifier Gates ⚠ ANTI-PATTERNS & FAILURE MODES × Alt-Text Noise & Mismatched Captions × OCR Hallucination on Low-DPI Scans × Biased Demographic & Cultural Corpora × Audio Background Noise & Crosstalk Leak	SigLIP-2 · OpenCLIP · DINOv2 Vision Native Early-Fusion Omni (GPT-4o / Gemini) Flow-Matching & DiT Diffusion Backbones Qwen2.5-VL · Pixtral · InternVL3 MLP / Perceiver / Q-Former Projectors 3D Spatio-Temporal Video RoPE RVQ Neural Audio Codecs (Mimi / SoundStream) Contrastive + Captioning Joint Pre-Train Zero-Shot ImageNet / VQA / ASR Evals 14x14 / 16x16 ViT Patch Embeddings Sigmoid Pairwise Contrastive Loss Classifier-Free Guidance (CFG) Dropout ⚠ ANTI-PATTERNS & FAILURE MODES × Modality Competition Degrading Text IQ × Bag-of-Patches Spatial Blindness × Quadratic ViT Token Explosion on 4K × Small-Text & Fine-Line Patch Blur
Embodied AI	Open X-Embodiment · DROID Robot Datasets Mind2Web · OSWorld · Rico GUI Traces Ego4D · Something-Something Video Logs ALOHA / VR Teleoperation Demonstrations DOM + Screenshot + Action Triplets 3D Point-Cloud & Depth-Map Alignment Proprioceptive Joint-State Normalization Success/Failure Trajectory Labeling Domain-Randomized Sim Data Generation RLDS / LeRobot Parquet Episode Format Coordinate Normalization [0, 1000] Camera Extrinsics & Intrinsics Calibration ⚠ ANTI-PATTERNS & FAILURE MODES × Teleoperator Hesitation & Suboptimal Demos × Coordinate Resolution Scaling Mismatch × Action-Space Embodiment Incompatibility × Unlogged Environment State Confounders	Vision-Language-Action (π0 · OpenVLA · RT-2) World Models (V-JEPA 2 · Cosmos · Genie 2) Diffusion Policy & Flow-Matching Actions Octo · CrossFormer Multi-Embodiment Base Latent Dynamics & Next-Frame Prediction 3D Point-Cloud & Voxel Encoders Discrete Action Bins vs. Continuous Heads Cross-Robot Morphology Tokenization Sim-to-Sim Zero-Shot Transfer Evals SE(3) Equivariant Pose Representations Video-to-Action Latent Conditioning Proprioceptive State History Window ⚠ ANTI-PATTERNS & FAILURE MODES × Causal Confusion (Spurious Correlations) × World Model Physics Hallucination × Multi-Modal Action Averaging Freeze × Camera Viewpoint Shift Fragility

ORGANISMS × STAGES	**5. SFT** Supervised Fine-Tuning, Instruction Templates, Sequence Packing & LoRA Adapters Instruction tuning, ChatML formatting, tool-call/reasoning trace SFT, loss masking, and PEFT (LoRA/DoRA). • Core Unit: Chat Templates, Packed Sequences & LoRA • Guide: Ch. 1.8, 3.2, 7.6, 9.4, 30.2 (SFT, Packing & PEFT Adapters)	**6. Post-Training** Preference Alignment (RLHF/DPO) & Verifiable-Reward Reinforcement Learning (RLVR/GRPO) Preference tuning (PPO, DPO, KTO) and critic-free Reinforcement Learning with Verifiable Rewards (RLVR, GRPO). • Core Unit: Preference Pairs, Rollouts & Advantages • Guide: Ch. 1.8, 3.2, 12.1–8, 30.2 (RLHF, DPO & RLVR / GRPO)
LLM	ChatML Templates & Completion-Only Masking HF TRL (SFTTrainer) · Axolotl · Unsloth PEFT Adapters: LoRA · QLoRA · DoRA LLaMA-Factory · Torchtune · Liger-Kernel Multipack Sequence Packing + Position IDs Tool-Use & Multi-Turn Trajectory SFT Cold-Start Chain-of-Thought Trace SFT MergeKit (SLERP / TIES / DARE Merging) IFEval Instruction-Following & Format Audit Mask Prompt Tokens (labels = -100) QLoRA 4-Bit NF4 + Double Quantization rsLoRA Rank-Stabilized Scaling $(\alpha/\sqrt{r})$ ⚠ ANTI-PATTERNS & FAILURE MODES × Training on Prompt Tokens (Missing -100) × Chat Template EOS/BOS Mismatch Loops × Cross-Sample Attention Bleed in Packing × Catastrophic Forgetting & Sycophancy	Preference Tuning: PPO · DPO · KTO · SimPO Critic-Free RLVR: GRPO · Dr. GRPO · DAPO verl · OpenRLHF · TRL (GRPO) · NeMo Aligner DeepSeek-R1 · OpenAI o3/o4 · QwQ-32B Verifiable Execution Oracles (Tests/Math) KL-Divergence & Policy Entropy Monitoring DAPO Clip-Higher & Dynamic Sampling Deliberative Alignment & Safety Rubrics Anti-Reward-Hacking & Length Control Group-Normalized Advantage A_i Ray + vLLM Distributed Async Rollouts Prime Intellect Verifiers Environments ⚠ ANTI-PATTERNS & FAILURE MODES × Reward Hacking & Length Inflation × Policy Entropy Collapse & Mode Collapse × GRPO Zero-Variance Advantage (All 0/1) × Sycophancy & Judge-Bias Exploitation
RAG	Cross-Encoder Reranker SFT (bge-reranker) Domain-Adapted Embedding Fine-Tuning Citation-Grounded QA SFT (RAFT) Retrieval-Augmented Fine-Tuning (RAFT) Distractor-Document Robustness Tuning Query-Rewriting & Decomposition SFT Structured JSON / SQL Query Generation SFT Hard-Negative Margin MSE Distillation Faithful Abstention ('I Don't Know') SFT Inline [Doc_i] Citation Span Formatting KL-Distillation from LLM Teacher Rankings LoRA Adapters on Bi-Encoder Backbones ⚠ ANTI-PATTERNS & FAILURE MODES × Memorizing Stale Facts Instead of Context × Ignoring Distractor Passages in SFT × Fabricated Citation Brackets ([1], [2]) × Overfitting Embedding to Narrow Queries	Search-R1: RLVR for Multi-Turn Search Citation Faithfulness & Groundedness DPO Abstention & Conflict-Resolution RL End-to-End Query-Retrieve-Reason RL Exact-Match / F1 Answer Verifier Rewards NLI Entailment Reward for Citations Penalizing Unsupported Extrapolation Retrieval Cost & Latency Penalty in RL RAGTruth & FaithBench Alignment Audits <search>query</search> Rollout Action Counterfactual Context Robustness Reward Hard-Negative Preference Pairs (Chosen/Rej) ⚠ ANTI-PATTERNS & FAILURE MODES × Reward Hacking via Redundant Searches × Cherry-Picking Weak Evidence to Please RM × Over-Abstention ('I Cannot Answer' Trap) × Hallucinating Plausible Uncited Claims
Multimodal	Visual Instruction Tuning (LLaVA-OneVision) GUI Grounding & Bounding-Box SFT Full-Duplex Speech-to-Speech SFT (Moshi) Frozen ViT + Trainable Projector + LLM SFT AnyRes High-Res Document/Chart SFT Interleaved Image-Text Multi-Turn SFT Voice Emotion, Prosody & Barge-In SFT Set-of-Mark (SoM) Visual Prompt SFT MMMU / ChartQA / DocVQA Eval Suites <image> / <audio> Special Token Alignment [x1, y1, x2, y2] Normalized Box Tokens LoRA on Vision Projector + LLM Layers ⚠ ANTI-PATTERNS & FAILURE MODES × Visual Hallucination (Answering Blindly) × Projector Collapse Ignoring Fine Details × Coordinate Token Off-by-One Drift × Voice SFT Losing Text Reasoning Ability	Multimodal RLVR (Visual Math & Geometry) Hallucination-Penalized VLM DPO (RLAIF-V) Diffusion DPO & Flow-GRPO Aesthetic RL OCR-Verified Chart & Table Reasoning RL Object-Detector Grounded Reward Oracles Voice Turn-Taking & Politeness Alignment Safety Refusal on Visual Jailbreaks Video Temporal Consistency RL Rewards MM-Vet / HallusionBench Alignment Evals IoU Bounding-Box Overlap Verifier Reward CLIP / PickScore Aesthetic Reward Models Contrastive Image-Conditioned DPO ⚠ ANTI-PATTERNS & FAILURE MODES × Text-Prior Shortcuts Ignoring the Image × Aesthetic Reward Hacking (Oversaturation) × Over-Refusal on Benign Medical/Art Images × Voice Interruption Hypersensitivity
Embodied AI	Behavioral Cloning (BC) & ACT Policies GUI Action-Trajectory SFT (OS-Atlas / Aguvis) Co-Fine-Tuning on Sim + Real Robot Demos Action Chunking with Transformers (ACT) Language-Conditioned Manipulation SFT Click(x,y) / Type(text) / Scroll GUI SFT Wrist + Third-Person Multi-View SFT LoRA Adaptation to New Robot Gripper Offline Action MSE & Task Success Rate k-Step Action Chunk Prediction Horizon Gaussian Mixture / Diffusion Action Head Proprioceptive History Dropout Regularizer ⚠ ANTI-PATTERNS & FAILURE MODES × Covariate Shift & Compounding BC Errors × Copycat Inertia Trap (Repeating Past Action) × Overfitting to Exact Background Lighting × GUI SFT Clicking Stale Static Coordinates	OSWorld / WebArena Verifiable GUI RLVR Sim-to-Real RL (PPO / SAC / HIL-SERL) Control-Barrier Safety Shielding RL DOM / Database State Verifier Rewards Human-in-the-Loop Sample-Efficient RL Domain Randomization (Friction/Mass/Lag) Residual RL on Top of Pre-Trained VLA Sparse Task-Completion vs. Dense Shaping Real-World Rollout Safety Reset Controllers Asymmetric Actor-Critic (Privileged Sim) Action Smoothness / Jerk Penalty Term Docker Snapshot Reset Between Rollouts ⚠ ANTI-PATTERNS & FAILURE MODES × Sim Physics Exploit (Glitching Simulator) × Destructive Real-World Hardware Exploration × GUI Agent Clicking 'Delete/Buy' for Reward × Sparse-Reward Credit Assignment Failure

ORGANISMS × STAGES	**7. Test-Time** Test-Time Compute Scaling, System-2 Chain-of-Thought Reasoning & Tree Search Inference-time System-2 scaling: Chain-of-Thought (<think>), Best-of-N, PRM reranking, and MCTS. • Core Unit: Thinking Tokens, Search Trees & Votes • Guide: Ch. 13.1–8, 15.1–9, 30.2 (Test-Time Compute & Search)	**8. Context** Context Engineering, Four-Tier Memory, Tool Calling, MCP & Skills Context budgeting, 4-tier agentic memory, function calling, Model Context Protocol (MCP), and SKILL.md. • Core Unit: Context Window, Memory & JSON-RPC • Guide: Ch. 17, 18.2–5, 22, 23 (Memory, Tools, MCP & Skills)
LLM	System-2 Chain-of-Thought (<think> CoT) Parallel Scaling: Best-of-N & Majority@k Sequential & Tree Search (PRM / MCTS) OpenAI o3 / Claude 4.6 Extended Thinking DeepSeek-R1 · Qwen3 Thinking · s1 Budget Process Reward Models (Step-Level PRM) Adaptive Thinking Budgets (Low/Med/High) Execution-Guided Code Filter before Vote Pass@k vs. Cons@k & AIME/ARC-AGI Evals s1 Budget Forcing ('Wait' Token Extension) PUCT Node Selection in Reasoning Trees Compute-Optimal Test-Time Allocation ⚠ ANTI-PATTERNS & FAILURE MODES × Overthinking & Analysis Paralysis (10x) × PRM Reward Hacking & Verifier Blindspots × Self-Correction Degrading Right Answers × Unfaithful Post-Hoc CoT Rationalization	Context Engineering & Token Budgeting Model Context Protocol (MCP / .mcp.json) 4-Tier Memory + Agent Skills (SKILL.md) Letta (MemGPT) · Mem0 · Zep · LangMem Working, Episodic, Semantic & Procedural Progressive Disclosure Skill Loading Sleep-Time Dreaming & Memory Consolidation Observation Masking & /compact Hygiene MCP OAuth 2.1 Scopes & LoCoMo Evals Prompt Caching Prefix Reuse (90% Savings) JSON-RPC 2.0 stdio & Streamable HTTP Keep Context Utilization < 65% Window ⚠ ANTI-PATTERNS & FAILURE MODES × Context Rot & Lost-in-the-Middle Decay × MCP Tool Bloat (100+ Schemas in Prompt) × Malicious SKILL.md & MCP Rug-Pulls × Memory Poisoning & Stale Fact Contradiction
RAG	Query Decomposition & Multi-Hop Planning Reciprocal Rank Fusion (Dense + BM25) CRAG & Self-RAG Critique-and-Retry Loops HyDE (Hypothetical Document Embeddings) Cross-Encoder Reranking Cascade (Top-100->10) Iterative IRCoT (Interleaved Retrieval + CoT) Minimum-Span Evidence Extraction Dynamic Top-k Cutoff via Confidence Score Latency-Bounded Multi-Index Speculation RRF Score = Σ 1 / (k + rank_i) Step-Back Prompting for Abstract Queries Cross-Document Contradiction Check ⚠ ANTI-PATTERNS & FAILURE MODES × Infinite Query-Rewrite Retrieval Loops × HyDE Hallucination Drifting Vector Search × Reranker Latency Blowing Up p99 SLA × Multi-Hop Error Propagation Across Steps	Qdrant · Milvus · pgvector · Weaviate GraphRAG · LightRAG · Neo4j Knowledge REFRAG & LLMLingua Context Compression Hybrid Dense + BM25 Index Orchestration Entity-Relation Knowledge Graph Communities Parent-Document & Window Context Expansion Time-Weighted & Recency-Biased Retrieval Document-Level RBAC & Tenant Isolation RAG Triad Evals (Faithfulness / Recall@k) HNSW ef_search & Metadata Pre-Filtering Leiden Community Summary Hierarchy KV-Cache Block Reuse for Static Docs ⚠ ANTI-PATTERNS & FAILURE MODES × Indirect Prompt Injection via Poisoned Docs × Cross-Tenant RBAC Data Leakage in Vector DB × Context Stuffing Drowning Signal in Noise × GraphRAG Entity-Extraction Cost Explosion
Multimodal	Visual Chain-of-Thought (Crop / Zoom / Pan) Active Perception & Region-of-Interest Scan Best-of-N VLM Verifier Reranking Iterative High-Res Crop Inspection (o3-Vision) Code-Execution Image Tooling (OpenCV/Matplotlib) Keyframe Selection + Dense Window Search Classifier-Free Guidance (CFG) Step Scaling Audio Re-Listening & Timestamp Verification Self-Consistency Over Multiple Crop Views Dynamic BBox Zoom-In Tool Call Denoising Diffusion Step Count (20-50) OCR-Assisted Visual Math Verification ⚠ ANTI-PATTERNS & FAILURE MODES × Zooming into Irrelevant Background Patches × Visual Confirmation Bias on Blurry Crops × Excessive High-Res Crops Blowing Token Cap × Diffusion Over-Stepping Artifact Burn-In	Adaptive ViT Token Merging (PruMerge / ToMe) Streaming Video KV-Cache Ring Buffers Cross-Modal CLIP / SigLIP Vector Memory Spatial-Temporal Video Token Compression Audio Transcript + Acoustic Embedding Memory MCP Image / Audio / Resource URI Handles OCR + Layout + Crop Tiered Context Cache Full-Duplex Voice Interruption State Tracking Video-MME Long-Video Needle-in-Haystack Drop 75% Redundant Background ViT Tokens Base64 vs. Presigned URI Media Payloads Sliding-Window Cross-Frame KV Eviction ⚠ ANTI-PATTERNS & FAILURE MODES × 10-Min Video Exhausting 128k Context Window × Aggressive Token Pruning Dropping Small Text × Base64 Image Bloat Crashing JSON-RPC Bus × Desynced Audio-Visual Timestamp Memory
Embodied AI	Model-Predictive Control (MPC) Rollouts World-Model Counterfactual Tree Search Visual Subgoal & Affordance Planning CEM / MPPI Trajectory Optimization in Sim VLM Failure Detection & Recovery Replan GUI Pre-Click Hover / DOM Verification 3D Value Map & VoxPoser Waypoint Search Receding-Horizon Action Chunk Execution Safety-Filter Critic Rejecting Unsafe Actions H-Step Imagination Rollout Horizon Collision-Free Configuration Space Check Speculative GUI Action Rollback ⚠ ANTI-PATTERNS & FAILURE MODES × Planning Latency Exceeding Control Rate × World-Model Rollout Diverging After 5s × Irreversible GUI Click Before Verification × Deadlock Oscillation Between Two Obstacles	3D Scene Graphs & Gaussian Splatting (3DGS) DOM + Accessibility Tree (AXTree) Pruning Episodic Spatial & Affordance Memory Semantic Voxel & OctoMap Occupancy Grids Set-of-Mark (SoM) Interactive Element IDs Proprioceptive & Force-Torque Ring Buffer Object Permanence & Tracked Entity State MCP Browser / OS / Robot Hardware Servers Spatial QA & Scene-State Consistency Evals Filter Invisible / Off-Screen DOM Nodes 6-DoF Object Pose Kalman Filter State Keyframe RGB-D Topological Map Nodes ⚠ ANTI-PATTERNS & FAILURE MODES × Raw 50,000-Node HTML DOM Overflowing Prompt × Dynamic Obstacle Ghosting in Stale 3D Map × Occlusion Forgetting (Out-of-Sight Vanishes) × Coordinate Frame Drift Over Long Episodes

ORGANISMS × STAGES	**9. Harness** Single-Agent Harness Engineering, Observe-Decide-Act Loops & Stateful Graph Workflows The CPU/OS of an agent: CLAUDE.md/AGENTS.md, Loop Engineering, Graph Engineering (LangGraph), and Harness^2. • Core Unit: State Transitions, Actions & Diffs • Guide: Ch. 16, 19, 20, 27 (Harness, Loops & Graphs)	**10. Multi-Agent** Multi-Agent Orchestration, Hierarchical Swarms, A2A Protocols & Shared State Multi-agent topologies (OWC), inter-agent protocols (A2A, ACP), parallel worktree swarms, and Agentic/Generative UI. • Core Unit: A2A Tasks, Worktrees & Swarm State • Guide: Ch. 1.14, 14, 24, 26, 28, 30.10 (A2A, MAS & Agentic UI)
LLM	Agent Harness (Prompt + Tools + State + Loop) Claude Code · Codex CLI · Cursor · OpenCode LangGraph · OpenAI Agents SDK · Harness^2 CLAUDE.md & AGENTS.md Repo Constitutions Observe-Decide-Act Loop & Stop Conditions Typed State Schema & Checkpointer Reducers Spec-Driven Development (Spec->Plan->Test) Decoupled Independent Verification (No Self-Grading) SWE-bench Verified/Pro & pass^k Reliability p^k Compounding Decay (0.95^20 = 35.8%) Git Worktree Checkpoints & Diff Rollback Human-in-the-Loop Destructive Tool Gate ⚠ ANTI-PATTERNS & FAILURE MODES × Agent Deleting / Mocking Failing Tests × Unbounded Loop Burn & Action Stagnation × God-Node Anti-Pattern & State Schema Bloat × CVE-2025-53773 IDE Config Auto-Approve RCE	Orchestrator–Worker–Critic (OWC) Pattern Google A2A Protocol · IBM ACP · ANP CrewAI · AutoGen / AG2 · LangGraph Swarm Hierarchical, Decentralized & Debate Swarms Parallel Git-Worktree Coding Subagents Independent Critic & Red-vs-Blue Verification CopilotKit · AG-UI · Vercel AI SDK Generative UI Agent Gateway (mTLS, OAuth OBO, FinOps Caps) Cross-Agent Cryptographic KYA & Audit Trails /.well-known/agent.json Capability Cards Per-Swarm Token & Dollar Circuit Breakers Blackboard Shared-Memory vs. Message Bus ⚠ ANTI-PATTERNS & FAILURE MODES × Cascading Hallucination & Swarm Echo Chamber × $O(N^2)$ Chatty Agent Token Explosion × Cross-Agent Confused Deputy & Privilege Escalation × Merge Conflicts Across Parallel Worktrees
RAG	Agentic RAG Workflows (LlamaIndex / Haystack) Plan-Search-Read-Synthesize-Verify Loop Deterministic Citation & Grounding Gate Adaptive Routing (No-RAG vs. Fast vs. Deep) Sub-Question DAG Execution & State Merge Schema-Enforced Structured Output Citations Source-Freshness & Authority Verification Fallback Web Search When Local KB Misses DeepResearch Bench & Citation Precision/Recall Max 3 Search Retries Before Escalation Span-Level Character Offset Provenance Read-Only Credentials for Search Connectors ⚠ ANTI-PATTERNS & FAILURE MODES × Search-Loop Thrashing on Rephrased Queries × Silent Fallback to Parametric Hallucination × Citation Laundering (Attributing False Claim) × Latency Blowup from Unbounded Deep Research	Federated Multi-Index Knowledge Brokers Parallel Deep-Research Subagent Swarms Cross-Domain Claim-Verification Critics Supervisor Routing Across SQL / Vector / Graph Map-Reduce Literature & Patent Synthesis Adversarial Fact-Checker vs. Writer Debate Per-Agent OAuth On-Behalf-Of (OBO) ACLs Shared Citation Blackboard & Deduplication Enterprise Cross-Silo Retrieval Benchmarks A2A Knowledge-Query Capability Discovery Provenance Hash Chain Across Subagents Token-Budget Allocation per Research Branch ⚠ ANTI-PATTERNS & FAILURE MODES × Subagent Circular Citation Laundering × Confused-Deputy Cross-Department Data Leak × 10 Subagents Fetching Identical Documents × Aggregator Dropping Minority Counter-Evidence
Multimodal	Voice Agent Harness (LiveKit / Pipecat) ComfyUI / Diffusers Iterative Gen-Critique Loop Visual-Render-Inspect-Fix Coding Loop VAD + Barge-In Interruption State Machine Speculative Turn-Taking & Backchanneling VLM Screenshot Diffing for Frontend UI Code Iterative Image Inpainting & ControlNet Loop C2PA Cryptographic Media Watermarking Word-Error-Rate (WER) & Turn-Latency Evals < 300ms End-of-Turn Speech Trigger Playwright Render -> VLM Critique -> CSS Fix Audio Echo Cancellation & Duplex Ducking ⚠ ANTI-PATTERNS & FAILURE MODES × Voice Agent Talking Over User (Barge-In Fail) × Echo Loop (Agent Transcribing Its Own TTS) × VLM Critique Loop Breaking Working Layout × Cascading ASR -> LLM -> TTS Latency (>3s)	Cross-Modal Specialist Production Swarms Director–Storyboard–Video–Audio Pipeline Multimodal Critic & Brand/Safety Gatekeeper Voice Concierge + Backend Tool Subagents Character-Sheet & Style Consistency Blackboard Parallel Scene Rendering + Audio Foley Sync VLM Visual QA Critic Grading Generated Media Streaming AG-UI Canvas & Live Media Artifacts End-to-End Multi-Modal Storyboard Evals Shared Latent Seed & IP-Adapter Reference WebRTC + A2A Split-Plane Media Streaming Automated Copyright & Likeness Guardrails ⚠ ANTI-PATTERNS & FAILURE MODES × Character Identity Drift Across Video Scenes × Voice Concierge Promising Actions Backend Fails × Audio-Visual Lip-Sync Desynchronization × Runaway Diffusion Rendering Costs in Swarm
Embodied AI	Computer-Use Harness (Operator / Claude CU) Hierarchical System-2 VLM + System-1 VLA Hardware Safety Interlocks & E-Stop Harness Screenshot -> Grounding -> Action -> Verify Loop Browser Pop-Up, Modal & Auth Recovery 2 Hz High-Level Subgoal + 50 Hz Motor Control Force/Torque Anomaly & Collision Reflex Stop Confirmation Gate Before Checkout / Send / Delete OSWorld / WebArena / AndroidWorld Benchmarks Post-Click DOM/Visual Diff Assertion Joint-Velocity & Workspace Bounding Box Automated Episode Reset & State Snapshot ⚠ ANTI-PATTERNS & FAILURE MODES × Visual Prompt Injection via Fake Screen Pop-Up × Clicking 'Submit/Pay' in Infinite Retry Loop × UI Loading Lag Causing Misaligned Double-Click × VLM Delusion Ignoring Physical Gripper Slip	Multi-Robot Fleet & Warehouse Orchestration Digital-Twin Sim + Physical Fleet Co-Execution Human–Robot–GUI Shared Autonomy Swarms MAPF (Multi-Agent Path Finding) Conflict-Free Heterogeneous Swarms (Drone + Rover + Arm) V2X Cooperative Perception & Spatial Sharing Cloud Fleet Planner + Local Edge Safety Reflex Operator Fleet Dashboard & Teleop Handoff Fleet Throughput, Deadlock & Collision Evals CBS (Conflict-Based Search) Spatio-Temporal DDS / ROS 2 Multi-Robot Namespace Isolation Graceful Degradation on Wi-Fi Drop ⚠ ANTI-PATTERNS & FAILURE MODES × Warehouse Corridor Deadlock & Traffic Gridlock × Shared-Map Poisoning from One Faulty Sensor × Network Partition Leaving Fleet Uncoordinated × Simultaneous Grasp Collision on Shared Object